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46 | BACK IN 1969, when Robert Noyce, having co-founded Intel barely a year ago, visited India to see if he could build a fabrication unit to make integrated circuits, no one had imagined a world dominated by electronics and semiconductor chips, as ICs are called. India offered him the opportunity to set up a fab, but it was too small for Intel's ambitions. Later, India did make a beginning with the Semiconductor Complex Ltd, but its facility in Mohali, near Chandigarh, was destroyed in a fire in 1989.

Today, in a post-Covid world, every major economy that has outsourced semiconductor chip manufacturing for decades, is seeking self-sufficiency. And so is India, with a ₹76,000-crore semiconductor and display fab scheme.

While Intel hasn't considered India yet for a fab, some leading names, such as the US-based Micron, Taiwan's Powerchip Semiconductor Manufacturing Corporation (PSMC), and Japan's Renesas have shown trust in the country's semiconductor ambitions. Micron is setting up an assembly, testing, marking, and packaging (ATMP) plant. PSMC and Renesas are working with Tata Electronics and CG Power on a semiconductor fab and a packaging plant, respectively.

But the government isn't stopping at fabs, testing and packaging plants: it is working on a 360-degree approach to creating an ecosystem and leading chip innovation.

The setting up of semiconductor fabs and ATMPs in any country usually leads to the development of a downstream ecosystem—industries that utilise the output to manufacture finished products. Putting in place such an ecosystem is critical to India's semiconductor dreams, says Ashwini Vaishnaw, Union Minister for Electronics and IT. "When semiconductor manufacturing begins in a country, many downstream industries start instantly, like laptop manufacturing, server manufacturing, electric vehicles, automobiles, machines used in factories, where semiconductors are used," he says.

For instance, South Korea's Simmtech, which makes printed circuit boards or PCBs, is coming to India to support Micron's semiconductor testing plant in Sanand, Gujarat, with an investment of ₹1,250 crore. "This is not our first time supporting our key customers. We did our project with Micron in China over 10 years ago. We have recently executed another project in Malaysia together. We have proven to the market that co-location investment boosted the growth of a semiconductor ecosystem in the region," says Simmtech CEO Jeffrey Chun.

Anticipating the needs of the downstream industry, Vaishnaw's ministry is open to introducing a scheme like SPECS or the Scheme for Promotion of Manufacturing of Electronic Components and Semiconductors. As Vaishnaw says, having manufacturers of wafer fabrication equipment or WFE is an important part of the India Semiconductor Mission. Industry experts say this is the most important success factor for a fab. WFE—large machinery employed in the production and processing of raw wafers to manufacture finalised chips—is the largest component of the expenditure on a fab.

PROJECTS APPROVED



- **Company Name:** Micron
- **Type:** ATMP
- **Location:** Sanand, Gujarat
- **Project Cost:** **\$2.7 billion** (₹22,000 crore)
- **Type of Chips:** Memory
- **Industry:** Electronics



- **Company:** CG Power
- **Type:** OSAT
- **Location:** Sanand, Gujarat
- **Project Cost:** **\$0.9 billion** (₹7,600 crore)
- **Type of Chips:** Packaging of legacy and advance chips
- **Industry:** Automotive, consumer, industrial, 5G

Danish Faruqui, CEO of Fab Economics, says, "With the entire world reeling under the constrained supply of WFE, which has a delivery lead time of multiple years in some cases, it is indeed existential for including semiconductor WFE manufacturing as part of India's decadal semiconductor road map."

He says India's WFE announcement was a masterstroke. Many across the world have cited India's entry into semiconductors as copying what China did 30 years ago, but India's approach is different and it factors in learnings from China's cardinal mistake.